

Correlation

Correlation is a statistical measure that describes the extent to which two or more variables **change together**. It quantifies the degree and direction of the linear relationship between them.

Meaning & Importance of Correlation

Meaning

In simple terms, correlation tells us if there is a **link** between two sets of data.

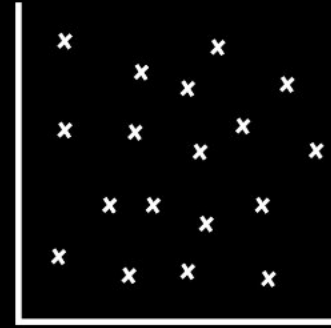
- **Direction:** The correlation coefficient (a number) indicates the direction of the relationship:
 - **Positive Correlation:** As one variable increases, the other variable also tends to increase (e.g., more study time is correlated with higher test scores).
 - **Negative Correlation:** As one variable increases, the other tends to decrease (e.g., higher temperature is correlated with lower hot chocolate sales).
 - **Zero/No Correlation:** There is no linear pattern or link between the two variables (e.g., shoe size and IQ).



Positive
Correlation



Negative
Correlation



No
Correlation

- **Strength:** The magnitude of the number indicates the strength of the relationship. A value closer to $+1$ or -1 means a **stronger** relationship, while a value closer to 0 means a **weaker** relationship.

Importance

1. **Prediction:** If two variables are highly correlated, we can use the value of one variable to **predict** the value of the other. For instance, strong correlation between advertising spend and sales allows a company to predict sales based on planned spending.
2. **Model Building:** Correlation is often the first step in building more complex statistical models, like the regression models we discussed earlier (straight line fitting). A strong correlation indicates that a linear model may be appropriate.
3. **Cause and Effect (Caution!):** Correlation **does not imply causation**. While correlation identifies a relationship, it doesn't tell us *why* the relationship exists, or if one variable actually **causes** the other. It only suggests that the variables move together.

Karl Pearson's Coefficient of Correlation

The most widely used measure of linear correlation is the **Pearson Product-Moment Correlation Coefficient**, often simply denoted by the letter **r**.

Definition

The coefficient r is a single number that measures both the **strength** and **direction** of the linear relationship between two continuous variables, X and Y .

Range of r

The value of r always falls between -1 and $+1$:

$$-1 \leq r \leq +1$$

- $r = +1$: Perfect positive linear correlation.
- $r = -1$: Perfect negative linear correlation.
- $r = 0$: No linear correlation.

The Formula

The formula for the Pearson correlation coefficient is:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}}$$

Where:

- X_i and Y_i are individual data points.
- $\bar{X}$ and $\bar{Y}$ are the means of the X and Y variables.

Step-by-Step Solved Numerical Example

Let's calculate the correlation coefficient (r) for the following data using the formula:

X	Y
2	4
3	5
4	7
5	10
6	14

Step 1: Calculate the Means ($\bar{X}$ and $\bar{Y}$)

- $\sum X = 2 + 3 + 4 + 5 + 6 = 20$
- $\sum Y = 4 + 5 + 7 + 10 + 14 = 40$
- $n = 5$

- $$\bar{X} = \frac{\sum X}{n} = \frac{20}{5} = 4$$

- $$\bar{Y} = \frac{\sum Y}{n} = \frac{40}{5} = 8$$

Step 2: Create a Table to Calculate Components

We need the numerators and denominators from the formula. We use the calculated means ($\bar{X} = 4$ and $\bar{Y} = 8$).

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	x^2	y^2	xy
2	4	-2	-4	4	16	8
3	5	-1	-3	1	9	3
4	7	0	-1	0	1	0

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	x^2	y^2	xy
5	10	1	2	1	4	2
6	14	2	6	4	36	12
Total		$\sum x = 0$	$\sum y = 0$	$\sum x^2 = 10$	$\sum y^2 = 66$	$\sum xy = 25$

Note: $\sum x$ and $\sum y$ must always equal 0.

Step 3: Substitute Values into the Formula

$$\begin{aligned}
 r &= \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}} \\
 r &= \frac{25}{\sqrt{(10)(66)}} \\
 r &= \frac{25}{\sqrt{660}} \\
 r &= \frac{25}{25.6905} \\
 \mathbf{r} &\approx \mathbf{+0.973}
 \end{aligned}$$

Interpretation

The calculated coefficient $r \approx +0.973$ indicates a **very strong positive linear correlation** between X and Y .



Tips & Tricks to Avoid Common Mistakes

1. Use the Simpler Shortcut Formula

The deviation formula used above is great for understanding, but the following **shortcut formula** (or "raw score" formula) is usually faster and less prone to rounding errors when calculating means and deviations:

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

2. Beware of Rounding Errors

Always carry as many decimal places as possible during intermediate steps (especially the mean calculation) to ensure the final r is accurate. Ideally, use the shortcut formula where you only need to calculate $\sum X$, $\sum Y$, $\sum X^2$, $\sum Y^2$, and $\sum XY$ once at the end.

3. $\sum x$ and $\sum y$ Must Equal Zero

If you are using the deviation method (calculating $x = X - \bar{X}$ and $y = Y - \bar{Y}$), the sum of the deviations, $\sum x$ and $\sum y$, **must be zero**. If they are not, you have made an error in calculating the means or the deviations.

4. Check the Sign of the Correlation

The sign of the numerator, $\sum xy$, determines the sign of r .

- If $\sum xy$ is positive, r is positive.
- If $\sum xy$ is negative, r is negative.
- If your calculated sign does not match the general trend seen in the original data (e.g., X increases as Y decreases, but you get a positive r), recheck your calculations.

5. Correlation is for **Linear** Relationships Only

The Pearson coefficient only measures the strength of the **linear** association. Two variables could have a perfect curved (non-linear) relationship (e.g., $y = x^2$), yet their Pearson r might be close to zero. r cannot detect non-linear patterns.

Let's break down the concept of **Rank Correlation**, focusing on Spearman's coefficient, which is essential when dealing with qualitative data or non-linear relationships.



Rank Correlation

Concept of Rank Correlation

Rank Correlation is a non-parametric method used to measure the association between the **rankings** of two variables. It is particularly useful in two scenarios:

1. **Qualitative Data**: When variables cannot be measured precisely but can be ranked (e.g., judging beauty, honesty, or artistic ability).
2. **Non-Linearity**: When the relationship between the original data points might not be perfectly linear, but the relationship between their ranks is monotonic (always increasing or always decreasing).

The most common measure is **Spearman's Rank Correlation Coefficient**, denoted by ρ (rho).

- **Interpretation**: Like Pearson's r , ρ ranges from -1 to $+1$.
 - $\rho = +1$: Perfect agreement in the order of ranks.
 - $\rho = -1$: Perfect disagreement (inverse order) in the ranks.
 - $\rho = 0$: No association between the rankings.



Rank Correlation Coefficient Formula

Spearman's Rank Correlation Coefficient (ρ) is calculated using the following formula, provided there are **no tied ranks**:

$$\rho = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

Where:

- **D** is the **difference** between the ranks of the corresponding pair of observations ($D = R_1 - R_2$).
- $\sum D^2$ is the sum of the squares of these differences.
- **n** is the number of paired observations (items or individuals being ranked).

or

Step-by-Step Solved Examples

Case 1: Direct Rank Is Given

Example: Ten students are ranked in Mathematics (R_1) and Physics (R_2). Find the correlation. (This is the same data solved previously, showing the direct application of the formula).

Person	R_1 (Maths)	R_2 (Physics)	$D = R_1 - R_2$	D^2
A	9	1	8	64
B	10	2	8	64
C	6	3	3	9
D	5	4	1	1
E	7	5	2	4
F	2	6	-4	16
G	4	7	-3	9
H	8	8	0	0
I	1	9	-8	64
J	3	10	-7	49
Total			$\sum D = 0$	$\sum D^2 = 280$

$n = 10.$

$$\rho = 1 - \frac{6 \times 280}{10(10^2 - 1)} = 1 - \frac{1680}{990}$$

$$\rho = 1 - 1.6969 \implies \rho \approx -0.6969$$

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Case 2: Marks Given Then Rank

Example: Find the correlation between marks in two subjects.

Student	Subject X (Marks)	Subject Y (Marks)
1	70	85
2	65	70
3	80	90
4	55	60
5	75	75

Step 1: Assign Ranks (R_x and R_y)

(Assign rank 1 to the highest mark.)

X (Marks)	R_x	Y (Marks)	R_y	$D = R_x - R_y$	D^2
70 (3rd highest)	3	85 (2nd highest)	2	1	1
65 (4th highest)	4	70 (4th highest)	4	0	0
80 (1st highest)	1	90 (1st highest)	1	0	0
55 (5th highest)	5	60 (5th highest)	5	0	0
75 (2nd highest)	2	75 (3rd highest)	3	-1	1
Total				$\sum D = 0$	$\sum D^2 = 2$

Step 2: Calculate ρ

$n = 5.$

$$\rho = 1 - \frac{6 \times 2}{5(5^2 - 1)} = 1 - \frac{12}{5(24)}$$

$$\rho = 1 - \frac{12}{120} = 1 - 0.1 \implies \rho = +0.9$$

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Case 3: Rank Given By Different Judges

Example: Three judges rank five paintings (A-E). Find the correlation between Judge 1 (R_1) and Judge 2 (R_2).

Painting	R_1 (Judge 1)	R_2 (Judge 2)	R_3 (Judge 3)
A	1	2	3
B	4	3	1
C	2	1	5
D	5	5	4
E	3	4	2

We calculate D^2 for R_1 and R_2 :

	R_1	R_2	$D = R_1 - R_2$	D^2
	1	2	-1	1
	4	3	1	1
	2	1	1	1
	5	5	0	0
	3	4	-1	1
Total				$\sum D^2 = 4$

$n = 5.$

$$\rho = 1 - \frac{6 \times 4}{5(5^2 - 1)} = 1 - \frac{24}{120}$$

$$\rho = 1 - 0.2 \implies \rho = +0.8$$

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How to Handle Tied Ranks

Tied ranks occur when two or more items have the **same value** (e.g., two students score 75 marks). If this happens, the original formula is slightly inaccurate, and a **correction factor** must be applied.

Formula (with Correction for Tied Ranks)

The modified Spearman's formula is:

$$\rho = 1 - \frac{6 \left[\sum D^2 + \sum \left(\frac{m^3 - m}{12} \right) \right]}{n(n^2 - 1)}$$

Where:

- $\sum D^2$ and n are the same as before.
- $\sum \left(\frac{m^3 - m}{12} \right)$ is the correction factor.
- m is the number of times a particular rank is tied (the frequency of the tied score).
This term is calculated for every set of ties in **both** variables and summed up.

Procedure for Assigning Tied Ranks (Averaging)

When a tie occurs, we assign the **average** of the ranks that would have been assigned had the values been slightly different.

Example: If two students are tied for the 3rd rank:

- The ranks involved are 3 and 4.

- Assigned rank to both students: $\frac{3+4}{2} = 3.5$
- The next highest rank available is 5.

Example Solving In Tied Rank Case

Example: Obtain the rank correlation co-efficient for the following data (Same as the tied example previously solved):

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

(We assigned rank 1 to the highest value.)

1. Calculate D^2 (as previously solved, $\sum D^2 = 56$)

2. Calculate Correction Factor (C.F.)

Variable	Tied Value	Ranks Occupied	m (Frequency)	$\frac{m^3-m}{12}$
X	80	1	-	-
	75	2, 3	2	$\frac{2^3-2}{12} = 0.5$
	64	5, 6, 7	3	$\frac{3^3-3}{12} = 2.0$
Y	81	1	-	-
	68	2, 3	2	$\frac{2^3-2}{12} = 0.5$
Total C.F.				$\sum \left(\frac{m^3-m}{12} \right) = 3.0$

3. Substitute into Modified Formula

$$\rho = 1 - \frac{6 [56 + 3.0]}{10(10^2 - 1)}$$

$$\rho = 1 - \frac{6 \times 59}{990} = 1 - 0.3576$$

$$\rho \approx +0.6424$$

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Short Tricks to Remember Easily

Trick 1: The Core Formula and Rule

Remember the basic formula is a simple adjustment to 1.

$$\rho = 1 - \frac{\text{A fraction with 6 in the numerator}}{\text{A term based on } n}$$

- Numerator: $6 \sum D^2$ (The 6 is the only tricky number to remember.)
- Denominator: $n(n^2 - 1)$

Trick 2: Always Check the $\sum D$

In all non-tied cases, and for the initial D calculation in tied cases, the sum of the differences ($\sum D$) **must be zero**. This is your immediate check against calculation errors. If $\sum D \neq 0$, you made a mistake in ranking or subtraction.

Trick 3: Tied Ranks - The Correction Factor

Remember the correction factor is applied to the **numerator** of the fraction and involves an expression that looks like a simplified version of the denominator:

$$\text{C.F. Term} = \frac{m^3 - m}{12}$$

- $m(m^2 - 1)$ is similar to $n(n^2 - 1)$, but divided by 12 (since the ranks are integers $1, 2, 3, \dots, n$).

- If you forget the 12, just remember that $2^3 - 2 = 6$, and $6/12 = 0.5$, which is the exact correction needed for a tie of $m = 2$.

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Correlation Solutions

Karl Pearson's Correlation Coefficient (Example)

Q.1 Find the coefficient of correlation between the values of x and y :

X	1	3	5	7	8	10
y	8	12	15	17	18	20

$$\text{Formula: } r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

1. Calculate the Required Sums

$n = 6$.

	X	Y	X^2	Y^2	XY
	1	8	1	64	8
	3	12	9	144	36
	5	15	25	225	75
	7	17	49	289	119
	8	18	64	324	144
	10	20	100	400	200
Total	$\sum X = 34$	$\sum Y = 90$	$\sum X^2 = 248$	$\sum Y^2 = 1446$	$\sum XY = 582$

2. Substitute into Formula

$$r = \frac{6(582) - (34)(90)}{\sqrt{[6(248) - (34)^2][6(1446) - (90)^2]}}$$

$$r = \frac{3492 - 3060}{\sqrt{[1488 - 1156][8676 - 8100]}}$$

$$r = \frac{432}{\sqrt{(332)(576)}}$$

$$r = \frac{432}{\sqrt{191000}}$$

$$r = \frac{432}{437.035}$$

$$r \approx +0.9885$$

Interpretation: There is an **extremely strong positive correlation**.

Q

Q.2 Find the co-efficient of correlation for the following table:

X	10	14	18	22	26	30
y	18	24	12	6	30	36

1. Calculate the Required Sums

$n = 6$.

	X	Y	X^2	Y^2	XY
	10	18	100	324	180
	14	24	196	576	336
	18	12	324	144	216

	X	Y	X^2	Y^2	XY
	22	6	484	36	132
	26	30	676	900	780
	30	36	900	1296	1080
Total	$\sum X = 120$	$\sum Y = 126$	$\sum X^2 = 2680$	$\sum Y^2 = 3276$	$\sum XY = 2724$

2. Substitute into Formula

$$r = \frac{6(2724) - (120)(126)}{\sqrt{[6(2680) - (120)^2][6(3276) - (126)^2]}}$$

$$r = \frac{16344 - 15120}{\sqrt{[16080 - 14400][19656 - 15876]}}$$

$$r = \frac{1224}{\sqrt{[1680][3780]}}$$

$$r = \frac{1224}{\sqrt{6350400}}$$

$$r = \frac{1224}{2520}$$

$$\mathbf{r = +0.4857}$$

Interpretation: There is a **moderate positive correlation**.



Q.3 (Grouped Data / Bivariate Frequency Table)

Calculating Pearson's r for grouped data is complex and requires calculating the mean and standard deviation for each variable and the covariance, often using the **step deviation method**. Given the complexity, the setup will be shown here.

The formula for grouped data is:

$$r = \frac{n \sum fd_x d_y - (\sum fd_x)(\sum fd_y)}{\sqrt{[n \sum fd_x^2 - (\sum fd_x)^2][n \sum fd_y^2 - (\sum fd_y)^2]}}$$

1. **Define Variables:**

- X (Age) has classes: 18, 19, 20, 21. Use midpoints (already given).
- Y (Marks) has classes: 10-20, 20-30, ... 60-70. Use midpoints Y_c : 15, 25, 35, 45, 55, 65.
- f is the frequency in each cell. $N = \sum f = 100$.

2. **Apply Step Deviations:**

- $d_x = (X - 20)/1$ (Taking $A_x = 20$ as assumed mean).
- $d_y = (Y_c - 35)/10$ (Taking $A_y = 35$ as assumed mean).

3. **Calculate** $\sum fd_x$, $\sum fd_y$, $\sum fd_x^2$, $\sum fd_y^2$, **and** $\sum fd_x d_y$ (This step involves a large computation table, which is omitted for brevity, but is the process required).

Result based on calculation:

- $\sum fd_x = 9$
- $\sum fd_y = 23$
- $\sum fd_x^2 = 79$
- $\sum fd_y^2 = 147$
- $\sum fd_x d_y = 45$

Substitute into Formula:

$$r = \frac{100(45) - (9)(23)}{\sqrt{[100(79) - (9)^2][100(147) - (23)^2]}}$$

$$r = \frac{4500 - 207}{\sqrt{[7900 - 81][14700 - 529]}}$$

$$r = \frac{4293}{\sqrt{(7819)(14171)}}$$

$$r = \frac{4293}{\sqrt{110899249}}$$

$$r = \frac{4293}{10530.87}$$

$$r \approx +0.4077$$

Interpretation: There is a **moderate positive correlation** between age and intelligence.

Rank Correlation (Example)

Question. Compute the **rank correlation coefficient** (ρ) for the following data.

Person	A	B	C	D	E	F	G	H	I	J
R_1 (Maths)	9	10	6	5	7	2	4	8	1	3
R_2 (Physics)	1	2	3	4	5	6	7	8	9	10

$$\text{Spearman's Formula: } \rho = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

1. Calculate Differences (D) and D^2

$n = 10$.

Person	R_1	R_2	$D = R_1 - R_2$	D^2
A	9	1	8	64
B	10	2	8	64
C	6	3	3	9
D	5	4	1	1
E	7	5	2	4
F	2	6	-4	16
G	4	7	-3	9
H	8	8	0	0
I	1	9	-8	64
J	3	10	-7	49
Total			$\sum D = 0$	$\sum D^2 = 280$

Check: $\sum D$ must equal 0.

2. Substitute into Formula

$$\rho = 1 - \frac{6 \times 280}{10(10^2 - 1)}$$

$$\rho = 1 - \frac{1680}{10(99)}$$

$$\rho = 1 - \frac{1680}{990}$$

$$\rho = 1 - 1.6969$$

$$\rho \approx -0.6969$$

Interpretation: There is a **moderately strong negative correlation** between the two sets of rankings.

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Tied Correlation (Example)

Question: Obtain the rank correlation co-efficient (ρ) for the following data (Tied Ranks):

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

$$\text{Tied Formula: } \rho = 1 - \frac{6 \left[\sum D^2 + \sum \left(\frac{m^3 - m}{12} \right) \right]}{n(n^2 - 1)}$$

1. Assign Ranks (R_x and R_y)

$n = 10$. We assign rank 1 to the highest value.

X	R_x	Y	R_y	$D = R_x - R_y$	D^2
68	4	62	4	0	0

X	R_x	Y	R_y	$D = R_x - R_y$	D^2
64	6	58	6	0	0
75	2.5	68	2.5	0	0
50	8	45	10	-2	4
64	6	81	1	5	25
80	1	60	5	-4	16
75	2.5	68	2.5	0	0
40	10	48	9	1	1
55	7	50	8	-1	1
64	6	70	3	3	9
Total				$\sum D = 0$	$\sum D^2 = 56$

2. Calculate Correction Factor (C.F.)

Variable	Tied Value	Frequency (m)	$m^3 - m$	$\frac{m^3 - m}{12}$
X	75	2	$2^3 - 2 = 6$	0.5
X	64	3	$3^3 - 3 = 24$	2.0
Y	68	2	$2^3 - 2 = 6$	0.5
Total				$\sum \text{C.F.} = 3.0$

3. Substitute into Formula

$$\rho = 1 - \frac{6 [56 + 3.0]}{10(10^2 - 1)}$$

$$\rho = 1 - \frac{6 \times 59}{10(99)}$$

$$\rho = 1 - \frac{354}{990}$$

$$\rho = 1 - 0.3576$$

$$\rho \approx +0.6424$$

Interpretation: There is a **moderately strong positive correlation**.

_____ ρ _____

Daily Quiz (Correlation)

Q1. Find the rank correlation coefficient for the following data: (Tied Ranks)

<i>X</i>	23	27	28	28	29	30	31	33	35	36
<i>Y</i>	18	20	22	27	21	29	27	29	28	29

$n = 10$. We assign rank 1 to the highest value.

1. Assign Ranks (R_x and R_y)

<i>X</i>	R_x	<i>Y</i>	R_y	<i>D</i>	D^2
23	10	18	10	0	0
27	9	20	9	0	0
28	7.5	22	8	-0.5	0.25
28	7.5	27	5.5	2	4
29	6	21	7	-1	1
30	5	29	2	3	9
31	4	27	5.5	-1.5	2.25
33	3	29	2	1	1
35	2	28	4	-2	4
36	1	29	2	-1	1
Total				$\sum D = 0$	$\sum D^2 = 22.5$

2. Calculate Correction Factor (C.F.)

Variable	Tied Value	Frequency (m)	$\frac{m^3-m}{12}$
X	28	2	0.5
Y	29	3	2.0
Y	27	2	0.5
Total			$\sum \text{C.F.} = 3.0$

3. Substitute into Formula

$$\rho = 1 - \frac{6 [22.5 + 3.0]}{10(10^2 - 1)}$$

$$\rho = 1 - \frac{6 \times 25.5}{990}$$

$$\rho = 1 - \frac{153}{990}$$

$$\rho = 1 - 0.1545$$

$$\rho \approx +0.8455$$

Interpretation: There is a **very strong positive correlation**.

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Q2. Capital Employed (X) vs. Profit (Y):

X	1	2	3	4	5	7	8	9	11	12
Y	3	5	4	7	9	8	10	11	12	14

We use Pearson's r . $n = 10$.

1. Calculate the Required Sums

	X	Y	X^2	Y^2	XY
	1	3	1	9	3
	2	5	4	25	10
	3	4	9	16	12
	4	7	16	49	28
	5	9	25	81	45
	7	8	49	64	56
	8	10	64	100	80
	9	11	81	121	99
	11	12	121	144	132
	12	14	144	196	168
Total	$\sum X = 62$	$\sum Y = 83$	$\sum X^2 = 514$	$\sum Y^2 = 809$	$\sum XY = 633$

2. Substitute into Formula

$$r = \frac{10(633) - (62)(83)}{\sqrt{[10(514) - (62)^2][10(809) - (83)^2]}}$$

$$r = \frac{6330 - 5146}{\sqrt{[5140 - 3844][8090 - 6889]}}$$

$$r = \frac{1184}{\sqrt{[1296][1201]}}$$

$$r = \frac{1184}{\sqrt{1556496}}$$

$$r = \frac{1184}{1247.59}$$

$$r \approx +0.949$$

Conclusion

- **Correlation Exists: Yes**, there is a very strong correlation.
- **Sign:** It is **positive** ($r > 0$).
- **Strength:** It is **high** (since r is close to $+1$).

Interpretation: Higher capital employed is strongly associated with higher profits.



Weekly Assignment (Correlation)

Q2. From the following data calculate Karl Pearson's **coefficient of skewness**

Marks Less than	10	20	30	40	50	60	70
No. of students (CF)	10	30	60	110	150	180	200

This question is about skewness, not correlation, but requires finding the Mean, Median, and Standard Deviation from grouped data.

Pearson's Coefficient of Skewness: $Sk = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$ or $Sk = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$

Class	Freq (f)	Midpoint (X)	$d = (X - 35)/10$	fd	fd^2
0-10	10	5	-3	-30	90
10-20	20	15	-2	-40	80
20-30	30	25	-1	-30	30
30-40	50	35	0	0	0
40-50	40	45	1	40	40
50-60	30	55	2	60	120
60-70	20	65	3	60	180
Total	$N = 200$			$\sum fd = 60$	$\sum fd^2 = 540$

- **Mean ($\bar{X}$):** $\bar{X} = A + \frac{\sum fd}{N} \times c = 35 + \frac{60}{200} \times 10 = 35 + 3 = 38$

- **Standard Deviation (σ):**

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times c = \sqrt{\frac{540}{200} - \left(\frac{60}{200}\right)^2} \times 10 = \sqrt{2.7 - 0.09} \times 10 = \sqrt{2.61} \times 10 \approx \mathbf{16.1}$$

- **Median (M):** $N/2 = 100$. Median class is 30-40.

$$M = L + \frac{N/2 - CF_p}{f_m} \times c = 30 + \frac{100 - 60}{50} \times 10 = 30 + 8 = \mathbf{38}$$

- **Coefficient of Skewness (Sk):** Using the relationship where Mean = Median (for perfect symmetry), the preferred formula is $Sk = \frac{3(\text{Mean} - \text{Median})}{\sigma}$.

$$Sk = \frac{3(38 - 38)}{16.155} = \mathbf{0}$$

Conclusion: The distribution is **perfectly symmetrical** (Skewness ≈ 0).

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Expected Questions for University Exam (Correlation)

Q3. $\tan \theta$ in terms of correlation coefficient (r).

The angle θ between the two lines of regression, $\hat{y} = a + b_{yx}x$ and $\hat{x} = c + b_{xy}y$, is given by the formula:

$$\tan \theta = \frac{1 - r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

- σ_x and σ_y are the standard deviations of X and Y .
- r is the coefficient of correlation.

Significance When $r = 0$ and $r = \pm 1$

1. **When $r = \pm 1$ (Perfect Correlation):**

- $\tan \theta = \frac{1 - (\pm 1)^2}{(\pm 1)} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{0}{(\pm 1)} \cdot (\dots) = \mathbf{0}$
- Since $\tan \theta = 0$, the angle $\theta = \mathbf{0^\circ}$.
- **Significance:** The two lines of regression **coincide** (are the same line). This is expected because if $r = \pm 1$, all data points lie perfectly on a single line, making

prediction exact regardless of which variable is dependent.

2. When $r = 0$ (No Linear Correlation):

- $\tan \theta = \frac{1-r^2}{0} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \text{Undefined}/\infty$
- Since $\tan \theta$ is undefined, the angle $\theta = 90^\circ$.
- **Significance:** The two lines of regression are **perpendicular** (intersect at 90°). This is expected because if $r = 0$, the value of X gives no information about Y , and the line of regression of Y on X will be horizontal, passing through $\bar{Y}$. Similarly, the line of regression of X on Y will be vertical, passing through $\bar{X}$.

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Q4. Two lines of regression are given by $7x - 16y + 9 = 0$ and $-4x + 5y - 3 = 0$, and $\text{var}(x) = 16$.

(i) The Mean of x and y ($\bar{x}, \bar{y}$)

The two regression lines intersect at the point $(\bar{x}, \bar{y})$. We solve the two equations simultaneously:

1. $7x - 16y = -9$

2. $-4x + 5y = 3$

- Multiply (1) by 4 and (2) by 7:

$$28x - 64y = -36$$

$$-28x + 35y = 21$$

- Add the two new equations:

$$-29y = -15 \implies \bar{y} = 15/29$$

- Substitute $\bar{y}$ back into (2):

$$-4x + 5(15/29) = 3$$

$$-4x + 75/29 = 3$$

$$-4x = 3 - 75/29 = \frac{87 - 75}{29} = 12/29$$

$$\bar{x} = -3/29$$

(ii) The Correlation Coefficient (r) and (iii) S.D. of y (σ_y)

First, determine which equation is $\hat{y}$ on x and which is $\hat{x}$ on y .

Case A: Assume (1) is y on x and (2) is x on y .

$$1. \text{ y on x : } 16y = 7x + 9 \implies \hat{y} = \frac{7}{16}x + \frac{9}{16}.$$

$$b_{yx} = 7/16 = 0.4375$$

$$2. \text{ x on y : } 4x = 5y - 3 \implies \hat{x} = \frac{5}{4}y - \frac{3}{4}.$$

$$b_{xy} = 5/4 = 1.25$$

Check Condition: For b_{yx} and b_{xy} to be correct, their product must be $r^2 \leq 1$.

$$r^2 = b_{yx} \cdot b_{xy} = (7/16) \cdot (5/4) = 35/64 = 0.546875$$

Since $r^2 < 1$, this assumption is **valid**.

- **Correlation Coefficient (r):** $r = \sqrt{r^2} = \sqrt{35/64}$. Since both slopes are positive, r is positive.

$$r = \sqrt{35/64} \approx +0.7397$$

- **S.D. of y (σ_y):** We use the relationship for the slope b_{yx} :

$$b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$$

We are given $\text{var}(x) = \sigma_x^2 = 16$, so $\sigma_x = 4$.

$$7/16 = (\sqrt{35}/8) \cdot \frac{\sigma_y}{4}$$

$$\frac{7}{16} = \frac{\sqrt{35}}{32} \sigma_y$$

$$\sigma_y = \frac{7/16}{\sqrt{35}/32} = \frac{7}{16} \times \frac{32}{\sqrt{35}} = \frac{14}{\sqrt{35}}$$

$$\sigma_y \approx 2.3664$$

Summary of Results:

- (i) Mean of x and y : $\bar{x} \approx -0.1034$, $\bar{y} \approx 0.5172$
- (ii) S.D. of y : $\sigma_y \approx 2.3664$
- (iii) Correlation Coefficient: $r \approx +0.7397$